FRC JAVA CODING LAB 7.0

FUTURE ARCHITECTURE REFERENCE

Non-authoritative package directions for large FRC robots

Document	Version	Status	Language
FAR-01	1.0	REFERENCE - NOT FROZEN	English

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REFERENCE ONLY - NOT A CONTRACT. This document does not modify Document A, Document B, Document C, AGENTS.md, or the Frozen Backbone. A listed package is never required merely because it appears here.

1. Purpose and Authority

This Future Architecture Reference (FAR) records conservative package directions that may help a large robot project grow without redesigning the frozen backbone. It is guidance for architecture review, not permission to create packages or classes.

If FAR conflicts with AGENTS.md or Documents A-C, the higher-authority document wins. English is the reference master; Vietnamese is an explanatory equivalent.

2. Frozen Backbone Preserved

CONTROL: Driver -> Xbox Controller -> controls -> commands -> subsystems -> io -> hardware

OBSERVATION: hardware -> IOInputs -> subsystem / estimator -> immutable Observation -> telemetry -> NT4 / Glass / log

FAR adds no stage to either flow. Future packages must fit an existing responsibility and preserve RobotContainer as the composition root.

3. Status Model

Status	Meaning	Creation rule
READY	Approved frozen location already available for real responsibilities.	Use now only when a lesson needs the responsibility; do not add empty subpackages.
RESERVED	Approved direction, intentionally absent until evidence exists.	Create only after the Creation Gate and Architecture Review pass.
DEFERRED	Direction is plausible but ownership or lifecycle is not mature.	Do not create in normal lessons; require a dedicated design review and migration plan.

4. Proposed Package Tree

```
frc.robot/
|- commands/
|   `- auto/                        [RESERVED]
|- controls/                        [READY]
|- subsystems/                      [READY]
|- io/                              [READY]
|   |- swerve/                      [RESERVED]
|   |- vision/                      [RESERVED]
|   `- replay/                      [DEFERRED]
|- observation/                     [READY]
|   |- swerve/                      [RESERVED]
|   |- vision/                      [RESERVED]
|   `- health/                      [DEFERRED]
|- estimation/                      [RESERVED]
|- telemetry/                       [READY]
|   |- logging/                     [RESERVED]
|   `- diagnostics/                 [RESERVED]
`- util/                            [READY]

src/main/deploy/pathplanner/
|- paths/                            [RESERVED]
`- autos/                            [RESERVED]
```

The tree reserves responsibility locations, not implementation. Empty packages, placeholder classes, and one-file abstractions are prohibited.

5. Responsibility Matrix

Package	Status	Allowed	Forbidden	Create when	Examples
commands/auto	RESERVED	Coordinate autonomous command sequences using subsystem APIs.	Hardware, pose ownership, estimation, vendor configuration.	Multiple autonomous routines share composition/registration policy.	PathPlanner routine coordination; named actions.
io/swerve	RESERVED	Vendor-isolated module, drive, encoder, and gyro hardware contracts.	Kinematics, pose estimation, NT4, autonomous decisions.	Swerve hardware has a reviewed IOInputs contract and real/sim plan.	Read module sensors; apply module outputs.
observation/swerve	RESERVED	Immutable swerve read models and pure evaluations.	Vendor types, mutable modules, control optimization, publishing.	Telemetry or logging needs stable interpreted swerve state.	Module validity, measured states, health meaning.
io/vision	RESERVED	Vendor-isolated camera or coprocessor transport and IOInputs.	Pose fusion, dashboard publishing, command decisions.	A real vision source and deterministic sim/noop contract exist.	Read target/pose candidates and connection facts.
observation/vision	RESERVED	Immutable vision measurements, validity, timing, and quality.	Camera access, NT topics, pose fusion side effects.	Consumers need a stable vendor-neutral measurement contract.	Timestamped pose candidate or target observation.
estimation	RESERVED	Cross-sensor state estimation that produces immutable Observations.	Hardware access, command scheduling, telemetry publishing, mechanism output.	Fusion spans mechanisms/sensors and no single subsystem can own it cleanly.	Robot pose, velocity, covariance, source validity.
telemetry/logging	RESERVED	Serialize immutable Observations to approved logs.	Control, hidden estimation, hardware reads, Observation mutation.	At least two mechanisms need consistent schemas/lifecycle and replay value.	Structured event and periodic observation recording.
io/replay	DEFERRED	Provide deterministic IOInputs from recorded data without hardware output.	Production motor commands, topic publication, silent time substitution.	Logging schema is stable and a replay verification objective is approved.	Offline regression or incident reconstruction.
telemetry/diagnostics	RESERVED	Present diagnostic interpretations from immutable Observations.	Safety interlocks, command cancellation, vendor polling.	Several mechanisms share diagnostic policy and stable reporting needs.	Fault summaries, connection reports, operator diagnostics.
observation/health	DEFERRED	Represent aggregate immutable robot-health meaning.	Active mitigation, device polling, mutable registry, scheduler actions.	Health semantics and owners are stable across multiple mechanisms.	Connectivity, configuration, thermal, power health summary.
deploy/pathplanner	RESERVED	Store approved PathPlanner paths and autos as deploy resources.	Java logic, subsystem state, hardware configuration.	PathPlanner is selected and autonomous verification is planned.	Paths and auto definition files.

6. Dependency Direction

Allowed direction: controls -> commands -> subsystems -> io -> hardware. A subsystem or estimator may produce immutable Observation values; telemetry, logging, diagnostics, and replay tooling may consume them.

commands/auto coordinates existing subsystem APIs. estimation consumes approved immutable inputs or copied values and produces immutable Observations. telemetry/logging and telemetry/diagnostics consume Observations only.

Forbidden direction: io -> NetworkTables or commands; observation -> hardware, telemetry, commands, controls, or mutable mechanism state; telemetry -> control; util -> mechanism-specific ownership; replay -> production hardware output.

7. Creation Gate

Create a RESERVED package only when all conditions are true: a current IN_PROGRESS lesson has a verified requirement; at least two cohesive responsibilities would otherwise be misplaced or one responsibility has clearly outgrown its current package; ownership and dependency direction are documented; the smallest viable API is reviewed; simulation and verification plans exist; and Architecture Review passes.

Do not create a package for anticipated complexity, naming symmetry, a temporary demo, one speculative class, or because another team uses it. DEFERRED packages additionally require a dedicated design review and recorded evidence that existing packages cannot safely own the capability.

8. Future Use Cases

Use case	Status	Direction
Swerve	RESERVED	Start with io/swerve; add observation/swerve only for stable interpreted state. Keep kinematics/control in the drive subsystem unless estimation genuinely becomes cross-cutting.
Vision	RESERVED	Transport belongs in io/vision, immutable measurements in observation/vision, and fusion in the owning subsystem or estimation.
Pose	RESERVED	Keep simple odometry in the drive subsystem. Create estimation only when multi-source fusion, uncertainty, and lifecycle justify independent ownership.
Autonomous	RESERVED	Use commands/auto and deploy/pathplanner. Autonomous coordinates; it does not own pose, hardware, or mechanism state.
Logging	RESERVED	Create telemetry/logging when consistent multi-mechanism schemas and lifecycle are required. Consume Observations only.
Replay	DEFERRED	Delay until stable logging schemas, deterministic time, and a verification objective exist. Replay must never actuate real hardware.
Diagnostics	RESERVED	Create telemetry/diagnostics for shared presentation and reporting. It may recommend attention but cannot command mitigation.
Health	DEFERRED	Keep mechanism health in mechanism Observations first. Aggregate only after shared semantics, validity, and ownership are proven.

9. Migration Roadmap

Stage	Status	Action	Exit evidence
0 - Current	READY	Use the frozen packages only. Preserve completed lessons.	No migration.
1 - Capability	RESERVED	Add mechanism IO and Observation subpackages only for verified hardware/features.	Build, simulation, architecture review.
2 - Coordination	RESERVED	Introduce commands/auto and deploy resources when multiple autonomous routines exist.	Autonomous verification and pose-source review.
3 - Estimation	RESERVED	Extract estimation only when multi-source fusion has independent ownership.	Determinism, timing, validity, covariance, replayable tests.
4 - Operations	RESERVED	Add logging and diagnostics when shared schemas/lifecycle are proven.	Schema review, storage/rate budget, read-only audit.
5 - Replay/Health	DEFERRED	Consider replay and aggregate health only after upstream contracts are stable.	Dedicated design review, migration plan, failure-safety evidence.

10. Final Review Checklist

[ ] FAR is labeled non-authoritative and REFERENCE only.

[ ] The frozen control and observation flows are unchanged.

[ ] Every future package is marked READY, RESERVED, or DEFERRED.

[ ] No RESERVED or DEFERRED package is required to exist.

[ ] Every package has purpose, allowed work, forbidden work, criteria, and examples.

[ ] No speculative class names or empty layers are proposed.

[ ] Swerve, Vision, Pose, Autonomous, Logging, Replay, Diagnostics, and Health are addressed.

[ ] RobotContainer remains composition-only.

[ ] IO and observation remain vendor-neutral at their public boundaries.

[ ] Subsystems/estimators produce Observations; telemetry only consumes them.

[ ] util remains generic.

[ ] Completed lessons and Documents A-C remain unchanged.

[ ] EN and VI have matching sections, statuses, matrices, and roadmap decisions.

[ ] DOCX and PDF layouts pass visual inspection.

Revision History

1.0 **|** 2026-08-01 **|** REFERENCE **|** Initial non-authoritative release.